Radical equations

1 Solve the following equations:

a $\sqrt{x} = \sqrt{13}$ b $\sqrt{h} = 5$ c $\sqrt{x} = \frac{3}{8}$ d $-3\sqrt{k} = -12$

2 Define a radical equation.

3 Is 4 a solution of the equation $\sqrt{x} = -2$?

4 Is 3 a solution of the equation $\sqrt{x-2} - 4 = \sqrt{x+6}$?

5 Is -2 a solution of the equation $\sqrt{x+18} = x+6$?

6 Solve the following equations:

a $\sqrt{y} - 6 = 1$

b $5\sqrt{x} + 3 = 28$

c $\sqrt{p} + 4 = 7$

d $\sqrt{y} + \frac{1}{5} = \frac{4}{5}$

e $\sqrt{x} - \frac{1}{4} = \frac{1}{3}$

f $\frac{\sqrt{x}}{-2} = -4$

g $5\sqrt{y} + 4 = 14$

h $-5\sqrt{c} - 2 = -17$

i $2\left(\sqrt{b} - 3\right) = 4$

j $\frac{\sqrt{a} - 1}{2} = 1$

k $\frac{\sqrt{b} + 1}{2} = \frac{3}{5}$

l $5\left(3\sqrt{a} + 8\right) = 70$

7 Solve the following equations:

a $\sqrt{x-5} = \sqrt{25}$

b $\sqrt{2x-6} = 2$

c $\sqrt{x-5} = 7$

d $\sqrt{x^2 - 8x + 16} = 7$

e $\sqrt{9x^2 - 48x + 64} = x$

f $\frac{12}{\sqrt{x-2}} = 3$

g $8\sqrt{x} - 4\sqrt{x} = 16$

h $\sqrt{3x-2} = 8$

i $\frac{\sqrt{x+9}}{2} - 8 = -2$

j $\frac{7\sqrt{x}}{11} + \frac{2\sqrt{x}}{11} = 9$

k $\sqrt{x^2 + 20} = 6$

8 Ray is given the equation $\sqrt{x+3}+7=4$ to solve. His work is shown:

a Does $x=6$ satisfy the equation $\sqrt{x+3}+7=4$?

b Describe the error Ray made in his work.

1. $\sqrt{x+3}+7=4$
2. $\sqrt{x+3}=-3$
3. $x+3=9$
4. $x=6$

9 Tom and Katrina are both given the equation $2\sqrt{x+71}-10=8$ to solve. Both have made errors in their work below:

Tom's work

1. $2\sqrt{x+71}-10=8$
2. $\sqrt{x+71}-10=4$
3. $\sqrt{x+71}=14$
4. $x+71=196$
5. $x=125$

Katrina's work

1. $2\sqrt{x+71}-10=8$
2. $2\sqrt{x+71}=18$
3. $\sqrt{x+71}=9$
4. $x+71=3$
5. $x=-68$

a Describe Tom's error.

b Describe Katrina's error.

c Solve the equation $2\sqrt{x+71}-10=8$.

10 Consider the equation $\sqrt{x+3}+10=0$. Without solving, explain how we can see that the equation has no real solutions.

11 Consider the equation $10x=1+3\sqrt{x}$.

a Solve the equation for x.

b Which value of x is an extraneous solution?

12 Consider the equation $\sqrt{x-5}=\sqrt{x}-1$.

a Solve the equation for x.

b Is this value of x an extraneous solution?

14 Solve the following equations:

a $\sqrt[3]{x+7}=4$

b $\sqrt[3]{3x-4}=2$

c $\sqrt[4]{x+2}+4=1$

15 Solve the following equations:

a $x^{\frac{3}{2}}=27$

b $x^{\frac{2}{7}}=4$

c $4x^{\frac{1}{4}}-3=9$

d $4x^{\frac{2}{7}}-1=15$

Applications

16 A square pyramid with a height of 12 units and a volume of V cubic units has a base with a side length of $b=\sqrt{\dfrac{V}{4}}$ units.

a Find the side length of the base of a square pyramid for each of the volumes in the table. Round your answers to the nearest tenth of a unit.

V	40	400	$4\,000$
b			

b Each volume in the table increased by a factor of 10. Did the corresponding side length of the base increase by a factor of 10?

17 An electron is constantly moving, and its position with respect to the fixed point ($x=0$) can be modeled by the equation $x=-\left(t-4\right)^2+4$, where x is its position after t seconds. After 4.5 seconds, it is at $x=3.75$.

a Solve the equation for t.

b Determine the time t at which the electron is at the fixed position.

c Determine how long it takes the electron to get from $x=-5.1$ to $x=-6.3$. Round your answer to the nearest tenth of a second.

18 The Beaufort Wind Scale was devised by British Rear Admiral Sir Francis Beaufort, in 1805 based upon observations of the effects of the wind. Beaufort numbers, B, are determined by the equation $B = 1.69\sqrt{s + 4.45} - 3.49$, where s is the speed of the wind in mph, and B is rounded to the nearest integer from 0 to 12.

a Using the table shown, find the Beaufort number of a speed of 40 mph.

b Classify the force of wind at a speed of 40 mph.

c In 1946, the scale was extended to accommodate strong hurricanes. A strong hurricane received a B value of exactly 15.
Determine the value of s to the nearest whole number.

d Any B values that round to 10 receive a Beaufort number of 10. Using technology, find an approximate range of wind speeds, to the nearest mile per hour, associated with a Beaufort number of 10.

	Beaufort Wind Scale
Beaufort number	Force of wind
0	Calm
1	Light air
2	Light breeze
3	Gentle breeze
4	Moderate breeze
5	Fresh breeze
6	Steady breeze
7	Moderate gale
8	Fresh gale
9	Strong gale
10	Whole gale
11	Storm
12	Hurricane